

Hyperparameter sensitivity tables (Tables 6–8)

Extension of manuscript Table 4 for one-at-a-time restoration hyperparameter sweeps. Metrics are mIoU and mTC (Section 4.2; mTC Eq. 17) after SAM 3 + causal inference on restored dashcam video.

Table 6

Restoration pipeline stage (manuscript Table 4)	Hyperparameters	mIoU (%)	mTC (%)
Raw input (no restoration)	—	65.4	77.2
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 220$	67.3	79.5
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 240$	68.6	81.0
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 260$	67.9	79.4

Table 6 — Sensitivity of flare threshold τ_{flare} (Section 3.2, Eq. 2) with Stage 1 flare suppression enabled and SAM 3 + causal segmentation held fixed. Other restoration stages are off for the Stage 1 block; bold row is the manuscript default ($\tau_{\text{flare}} = 240$, Table 4).

Table 7

Restoration pipeline stage (manuscript Table 4)	Hyperparameters	mIoU (%)	mTC (%)
Raw input (no restoration)	—	65.4	77.2
+ Stage 2: DCP dehazing (guided)	$\sigma = 2.0$	70.2	83.2
+ Stage 2: DCP dehazing (guided)	$\sigma = 3.0$	71.4	84.6
+ Stage 2: DCP dehazing (guided)	$\sigma = 4.0$	70.6	83.0

Table 7 — Sensitivity of Gaussian feather scale σ (Section 3.2, Eq. 3) with Stages 1–2 active ($\tau_{\text{flare}} = 240$ fixed). Bold row matches manuscript Table 4 ($\sigma = 3.0$).

Table 8

Restoration pipeline stage (manuscript Table 4)	Hyperparameters	mIoU (%)	mTC (%)
Raw input (no restoration)	—	65.4	77.2
+ Stage 3: CLAHE & gamma	$\gamma = 0.8$	71.2	85.9
+ Stage 3: CLAHE & gamma	$\gamma = 1.4$	73.1	86.8
+ Stage 3: CLAHE & gamma	$\gamma = 2.2$	71.6	86.2

Table 8 — Sensitivity of gamma γ (Section 3.2, Eq. 10) with Stages 1–3 active ($\tau_{\text{flare}} = 240$, $\sigma = 3.0$ fixed). Bold row matches manuscript Table 4 ($\gamma = 1.4$). Full tri-stage result: 74.8% mIoU / 88.4% mTC (Table 4, final row).